
STC 5-Week Mentorship 3.0

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STC Data Analyst Project – Week 3 Report (FINANCE INDUSTRY)

INTRODUCTION

This report presents an analysis of customer complaint data using data analysis and visualization techniques. The aim of the analysis was to identify complaint trends, understand customer behavior patterns, evaluate company response performance, and uncover key business insights that can support better decision-making.

Interactive dashboards and visual reports were developed using Microsoft Power BI to transform raw data into meaningful and easy-to-understand insights. Microsoft PowerPoint was also used to design the dashboard wireframe and layout structure before development.

DATASET OVERVIEW

The dataset contains consumer complaints related to financial products and services.

- Number of rows before cleaning: 62,516
- Number of columns before cleaning: 12

Each record represents an individual complaint submitted by a customer and includes details such as product category, complaint issue, submission channel, response type, response status, and complaint dates.

The dataset was suitable for analyzing customer dissatisfaction patterns, service performance, and operational efficiency in handling complaints.

BUSINESS QUESTIONS

The analysis focused on answering the following business questions:

- i. What are the most common customer complaint issues?
- ii. Which products receive the highest number of complaints?
- iii. Which complaint submission channels are most frequently used?
- iv. Which states record the highest number of consumer complaints?
- v. How have consumer complaints changed over time?
- vi. What are the most common company responses to complaints?
- vii. How effective are companies in providing timely responses?
- viii. Which products have the highest rate of untimely responses?

TOOLS USED

Microsoft Power BI

- It was used for data analysis, visualization, and dashboard development.
- Import and organize the cleaned dataset
- Create interactive dashboards and visual reports
- Build KPI cards for: total complaints, affected products, timely response rate
- Create visualizations such as: bar charts, column charts, line charts, doughnut charts
- Analyze complaint trends, products, states, and response performance
- Apply filters and slicers for interactivity
- Create DAX measures and calculations for analysis

Microsoft PowerPoint

- It was used for dashboard planning and wireframe design.
- Create dashboard wireframes and mockups
- Plan the arrangement of charts, KPIs, and filters
- Design the dashboard layout and storytelling structure
- Test color combinations using the color picker tool
- Ensure proper alignment and spacing of dashboard components

ANALYSIS PERFORMED

Several steps were carried out during the analysis process:

- The dataset was cleaned and prepared for analysis.
- Missing values and inconsistencies were identified and corrected.
- Interactive dashboards and visual reports were created in Power BI.
- DAX measures were developed to calculate metrics such as timely response rates.
- Charts and visuals were built to analyze complaint trends, complaints by product, complaints by state, complaint submission channels, and company response types.
- Interactive filters for product, state, and year were added to improve dashboard usability and exploration.
- Dashboard layouts were carefully designed to improve storytelling and business interpretation.

KEY INSIGHTS

i. Complaint Volume

A total of 62.52K customer complaints were recorded in the dataset.

ii. Most Affected Product

The Checking or Savings Account category recorded the highest number of complaints, with approximately 24.81K complaints, followed by Credit Card or Prepaid Card complaints.

iii. Timely Response Performance

Companies maintained a strong customer service performance with a 93.77% timely response rate, showing that most complaints were resolved within the expected timeframe.

iv. Complaint Trends

Customer complaints increased steadily from 2017 and reached their peak around 2022 before declining slightly in 2023.

v. Common Customer Issues

The most reported issue by customers was managing an Account, followed by Incorrect Information on Reports and Problems with Purchases.

vi. Complaint Submission Channels

Most complaints were submitted through the web channel, indicating that customers prefer digital communication channels over phone calls or emails.

vii. Geographic Distribution

California recorded the highest number of complaints, followed by Florida, Texas, and New York.

viii. Company Response Types

Most complaints were resolved through explanations provided by companies, while fewer complaints resulted in monetary or non-monetary relief.

DASHBOARD & VISUALS

The Power BI dashboard contains two pages designed to provide a clear understanding of customer complaints and company performance.

Page 1: Complaint Overview

This page includes:

- Total complaints KPI
- Number of affected products
- Timely response rate
- Complaint trends over time
- Complaints by product
- Most common complaint issues
- Interactive product filter
- Interactive state filter

Page 2: Company Performance

This page includes:

- Company response types
- Timely vs untimely responses
- Complaints by submission channel
- Complaints by state
- Interactive year filters

The dashboards were designed to present insights clearly while making it easier for users to explore customer behavior, complaint trends, and company response performance interactively.

ALIGNMENT WITH BUSINESS QUESTIONS

- Yes

The analysis successfully addressed the key business questions by identifying major complaint trends, customer behavior patterns, response performance, and the geographic distribution of complaints.

However, the dataset provided could not fully answer all the business questions generated during Week 1 of the project. As a result, additional business questions were developed to ensure the analysis remained relevant and insightful based on the available data.

ADDITIONAL NOTES

Challenges Faced

Some of the challenges encountered during the project included:

- Managing long category names within dashboard visuals
- Maintaining proper alignment and spacing across dashboard components
- Ensuring consistency in visual design and dashboard storytelling

Assumptions Made

Timely response calculations were based on the response status information available in the dataset.

Additional Highlights

The dashboard includes interactive filtering options that allow users to explore complaint patterns by year, state, and product category, making the analysis more flexible and insightful for decision-making.